DUONG Quentin

Accelerator Physicist (PhD)
&
Nuclear Engineer

PROFILE

+33 (0)7.60.84.11.91

q.duong13@gmail.com

linkedin.com/in/duong-q

duongque.github.io

French driving licence

TOOLS & CODE

PYTHON

C / C++

VBA

POWERSHELL

BASH

LABVIEW

COMSOL

FLUKA

LANGUAGES

FRENCH – *Native language*

ENGLISH – C1

MANDARIN – A2

SPANISH – A2

HOBBIES

CINEMA

COOKING

MARTIAL ARTS

PHYSICS

WORK EXPERIENCE

R&D Engineer – Automation & Data

Jan 2026 – Today
Freelance (Paris – France)

Design of an end-to-end pipeline for processing unstructured documents (PDF): assisted information extraction, automated structuring and integration into a traceable data system.

➤ *Python · Data pipeline · LLM API · Document automation · Traceability*

Accelerator Physics Researcher

Oct 2022 – Sep 2025
CERN (Genève – Suisse)

3-year doctoral program on the measurement and characterisation of electron cloud in the LHC. Regular work in radioactive environments with precision measurement instruments. Daily monitoring of LHC operation and participation in CLEAR runs – a 200 MeV electron linac at CERN used for FLASH-VHEE research.

➤ *Precision calibration · Instrumentation in radioactive environments · Data acquisition & analysis (7000+ fills) · PyECLOUD · Multiphysics simulation COMSOL · Monte Carlo radiation transport code FLUKA*

Particle Physics Researcher

Apr 2022 – Aug 2022
KEK (Tsukuba – Japan)

5-month internship on cosmic muon measurements in an underground cavern with the COMET experiment at J-PARC. Muons used as reference particles for dosimetric calibration.

➤ *Gaseous detector · Dosimetric calibration · Long-duration measurements (C++) · Advanced statistical analysis*

EDUCATION

PhD | Highest Honours

Oct 2022 – Dec 2025
Paris-Saclay University (Paris – France)

Physics of Particle Accelerators
Thesis: Electron cloud study with the CERN LHC Vacuum Pilot Sector

➤ *Particle accelerator technology · Proton beam dynamics · Vacuum systems · Electron-cloud dynamics · Beam-matter interactions · PyECLOUD · Machine Learning · Monte Carlo radiation transport code FLUKA*

Master Degree | High Honours

Jan 2022 – Mar 2022
JUAS (Archamps – France)

Physics & Technology of Particle Accelerators

➤ *Accelerator design · Beam dynamics & instrumentation · Collective effects · Normal & superconducting magnets · Synchrotron radiation · Cryogenics · RF Engineering · Vacuum systems*

Oct 2020 – Dec 2021
Grenoble Alpes University (Grenoble – France)

Nuclear Physics

➤ *Advanced nuclear physics · Plasma physics · Nuclear fusion*

Engineer Degree | Honours

Oct 2019 – Aug 2022
Grenoble INP – PHELMA (Grenoble – France)

Nuclear Engineering & Physics

➤ *Reactor physics · Nuclear physics · Beam-matter interactions & Dosimetry · Radiation protection · Neutron transport simulation (Fortran, MCNP) · Thermohydraulics · Fluid mechanics*

Bachelor Degree | Honours

Oct 2019 – Aug 2020
Grenoble INP – PHELMA (Grenoble – France)

Physics, Electronics & Telecomunications

➤ *Electronics · Signal treatment · Automatic control & Microprocessors · Material science · Network protocols · Computer science · Physics · Mathematics*

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- PYTHON
- C / C++
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- BASH
- LABVIEW
- COMSOL
- FLUKA

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- **FRENCH** – *Native language*
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HOBBIES

- CINEMA
- COOKING
- MARTIAL ARTS
- PHYSICS

TRAINING

Topical course on Medical Accelerators

Jun 2026
CAS (Riga – Latvia)

11-day intensive course organised by the CERN Accelerator School (CAS) in collaboration with Riga Technical University, covering medical applications of particle accelerators

- *Medical accelerators · Radiobiology & Oncology · Radioisotope production · Nuclear Medicine · Medical imaging (PET/SPECT) · FLASH-VHEE · Radiotherapy · Regulatory framework*

Challenge-Based Innovation on Particle Accelerators for Healthcare

Jun 2025 – Jul 2025
iFast (Archamps – France)

10 intensive days on medical innovation through particle accelerators. Complete technical proposal in a multidisciplinary team: "Fibrosis Disruption: SMITE Protocol for FLASH RT-Activated Anti-Fibrotic nanoparticles".

- *Heavy ion radiobiology · Medical Linacs & Dosimetry · Radionuclide production · Accelerator physics & technology · Science mediation*

PAPER & TALK

Electron cloud energy distribution measurement and model in the LHC with theVacuum Pilot Sector

Q. Duong, V. Baglin, G. Sattonnay
Under Reviewing – Physical Review Accelerator and Beams

In situ estimation of maximum Secondary Electron Yield in the LHC Vacuum Pilot Sector beam pipe via electron cloud measurements and numerical simulations

Editors' Suggestion

Q. Duong, V. Baglin, G. Sattonnay
Août 2026 – Physical Review Accelerator and Beams, Vol 29, 083503

SEY estimation with VPS electron cloud measurements and PyECLOUD simulations

CERN – TE/VSC Seminar
Apr 2025 – Meyrin, Switzerland

Some bulk Cu and aC coating preliminary results obtained with the Vacuum Pilot Sector during 2023 run

Joint Accelerator Performance Workshop
Dec 2023 – Montreux, Switzerland